

Sushanth Buddhala

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Fairfax, Virginia

PROFILE

Data Analytics Engineering graduate student with hands-on project experience building RAG pipelines, agentic AI workflows, and machine learning models. Skilled in Python, SQL, Power BI, and the OpenAI API, with a focus on designing prompt engineering workflows, implementing data validation, and translating complex datasets into actionable insights.

EDUCATION

George Mason University

Virginia, USA

M.S. in Data Analytics Engineering

- **Relevant Coursework:** *Analytics & Big Data, Intro to Analytics & Modeling, Principles of Data Management & Mining, Visualization for Analytics, Big Data Essentials, Data Mining for Business Analytics, Interpretable Machine Learning, Data Analytics Project*

Gokaraju Rangaraju Institute of Engineering and Technology

Telangana, India

B.S. in Computer Science Engineering

- **Relevant Coursework:** *Linear Algebra Differential Calculus, Programming for Problem Solving, Differential Equation and Vector Calculus, Database Management System, Python Programming, Artificial Intelligence, Data Analytics with Python, Machine Learning, Big Data Analytics, Cloud Computing, Data Warehousing and Data Mining*

TECHNICAL SKILLS

Languages & Tools: Python, SQL, R, DAX, Microsoft Excel, Databricks, Snowflake, Alteryx, Duck DB, Pydantic, Pytest, FastAPI

Data & ML: Data Analysis, Data modeling, Data Warehousing, Data visualization, Data validation, ETL Pipelines, Machine Learning, Unsupervised learning, Data pipelines, Product Analysis

Visualization & Cloud: Power BI, AWS

AI & LLM: Cursor.ai, Claude code, Multi- Agent Systems, Agent Orchestration, Large Language Models, LLM Observability, LLM Tools, Model Evaluation, RAG pipelines, Agentic AI

PROJECT EXPERIENCE

Housing Migration Analytics Platform (SQL + RAG)

Nov 2025 – Jan 2026

- Designed and implemented a normalized SQLite analytics database containing 395K+ U.S. housing, migration, AHS, and CHAS records, leveraging Cursor AI to accelerate development and optimize indexed tables for analytical querying.
- Built a schema-aware, schema-sensitive RAG pipeline using OpenAI embeddings to generate safe, explainable SQL, preventing hallucinations, destructive queries, and invalid SQL.
- Improved query accuracy and reliability by enforcing schema validation and SQL guardrails, enabling non-technical users to safely explore data.

Predictive Pavement Condition Analysis (ML & Big Data)

Jan 2025 – May 2025

- Processed and analyzed 2,000+ pavement condition records using imputation, normalization, encoding, and outlier handling to improve data quality.
- Applied K-Means clustering to identify three pavement deterioration patterns, revealing faster degradation in high-traffic environments.
- Built and evaluated regression models (Gradient Boosting, Ridge, Lasso), achieving best performance with Gradient Boosting (RMSE $\approx$ 890).

Obesity Level Analysis (Statistical & Machine Learning Study)

Aug 2024 – Nov 2024

- Conducted exploratory data analysis on 2,000+ records to uncover relationships between BMI, lifestyle factors, and family history.
- Developed and validated Logistic Regression, Decision Tree, SVM, and Clustering models, achieving up to 85% accuracy (best with SVM).
- Identified diet, physical activity, and family history as key predictors, translating findings into actionable public-health insights.

PROFESSIONAL CERTIFICATES

• IBM Data Analyst Professional Certificate by Coursera • AWS Academy Graduate - Cloud Foundation • Generative AI for Data Analytics by Coursera